ELS2T: Efficient Lightweight Spectral–Spatial Transformer for Hyperspectral Image Classification

Shichao Zhang, Jiahua Zhang, Xiaopeng Wang, Jingwen Wang, and Zhenjiang Wu

***Abstract*— In recent years, convolutional neural networks (CNNs) have been extensively used in hyperspectral image (HSI) classification tasks and achieved desirable performance. However, CNNs expand the receptive field by stacking convolutional layers and pooling layers, but the actual receptive field is still insufficient, so it is hard to capture the global representations of HSIs. In addition, the existing CNN models used for HSI classification have low-computational efficiency, and cannot effectively fuse spectral and spatial information. To alleviate the above limitations, this article proposes a novel efficient lightweight spectral–spatial transformer (ELS2T) specially designed for HSI classification. The transformer model can model the long-distance feature dependencies in the HSI cube. First, we design a global multiscale attention module (GMAM) to effectively highlight the useful information and weaken the useless information. Second, considering the different importance of spectral and spatial information for classification tasks, an adaptive feature fusion module (AFFM) is proposed to adaptively fuse the acquired spectral and spatial information. Finally, to improve the computational efficiency, we design the lightweight separable spatial–spectral self-attention (S^3A) module to replace the multihead self-attention (MHSA) module in the transformer encoder. Experimental results on the four well-known hyperspectral datasets show that our model is superior to the other state-of-the-art deep learning (DL) methods in both computational efficiency and classification performance.**

***Index Terms*— Adaptive feature fusion, deep learning (DL), efficient lightweight spectral–spatial transformer (ELS2T), global multiscale attention, hyperspectral image (HSI) classification, self-attention mechanism.**

I. Introduction

HYPERSPECTRAL images (HSIs) have the property of spectrum integration, which can simultaneously record the spectral information and spatial information of observation objects. In addition, HSIs have the characteristics of high imaging spectral resolution and fast nondestructive testing [1]. Thus, HSIs are widely used in many fields, such as crop classification and yield estimation [2], mineral resource exploration [3], urban land planning [4], marine monitoring [5], fire prevention, and postdisaster assessment [6]. HSI classification is a key step in analyzing HSI, which refers to identifying the category of each pixel in the HSI and has attracted considerable attention.

In the early stages of HSI classification, researchers generally design features manually and then classify them using classic machine learning methods such as K-nearest neighbors [7], random forest [8], and support vector machine [9]. However, these methods usually extract fixed pattern features from hyperspectral data, and the extraction process depends on prior knowledge and appropriate parameter settings. These limitations seriously affect the robustness and discrimination of the extracted features, making it difficult to obtain satisfactory results in complex scenes.

More recently, with the continuous development of deep learning (DL) technology, it has been widely used in computer vision (CV) tasks, for instance, image dehazing [10], [11] and low-light image enhancement [12]. Subsequently, it has also stimulated the development of DL in the field of remote sensing image processing, such as its extensive use in tasks such as semantic segmentation [13], [14], [15], [16], [17], scene classification [18], [19], and change detection [20], [21], [22]. In HSI classification, DL-based methods that were initially tried mainly include stacked autoencoders (SAEs) [23], deep belief neural networks (DBNs) [24], and recurrent neural networks (RNNs) [25]. However, these methods use a 1-D vector as input and only consider the spectral information of pixels in HSI. Although the spectral resolution of HSI is very high, the existence of "different spectra of the same substance" and "foreign matters with the same spectrum" requires that HSI classification methods should jointly obtain spectral and spatial information to effectively improve the classification accuracy. Later, some advanced DL models are used as the backbone of HSI classification, such as graph neural networks (GNNs), [26], [27], [28], and fully convolutional networks (FCNs), [29], [30]. Due to their superiority in processing image data, convolutional neural networks (CNNs) have become the mainstream method of HSI classification. Hybrid spectral CNN (HybridSN) [31] combines 2-D convolution and 3-D convolution to explore spectral and spatial features. Nie et al. [32] introduced deformable convolution into the network. Residual learning effectively avoids gradient disappearance and explosion, and has been widely used in HSI classification models. Song et al. [33] introduced residual connection and fuse features at different levels.

Manuscript received 21 March 2023; revised 13 June 2023; accepted 24 July 2023. Date of publication 27 July 2023; date of current version 10 August 2023. This work was supported in part by the Central Guiding Local Science and Technology Development Fund of Shandong-Yellow River Basin Collaborative Science and Technology Innovation Special Project "Study on the Prediction Technology and Countermeasures of Compound High Temperature and Drought in Ningxia" under Grant YDZX2023019, in part by the Natural Science Foundation of Shandong Province under Grant 2018GNC110025 and Grant ZR2020QF067, in part by the "Taishan Scholar" Project of Shandong Province under Grant TSXZ201712, and in part by the National Natural Science Foundation of China under Grant 41871253. *(Corresponding author: Jiahua Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2023.3299442

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Zhong et al. [34] constructed spectral and spatial residual blocks to extract spectral–spatial features. Different targets in HSI have different sizes, so researchers consider multiscale feature extraction. Yang et al. [35] designed a parallel multipath network (PMN) to extract enough multiscale features. Zhang et al. [36] considered extracting multiscale spectral and spatial features, respectively. The attention mechanism originates from the study of human vision, many researchers have embedded attention modules in the model and achieved significant accuracy improvement. Shi et al. [37] proposed a dual-branch network with attention modules. Zhang et al. [38] introduced attention modules and gave different weights to multiscale features. Zhang et al. [39] considered the similarity between the central pixel of HSI cubes and the surrounding pixels, so they designed a novel attention module named homogeneous pixel detection module (HPDM). Yang et al. [40] introduced a cross-attention spectral–spatial network (CASSN) to alleviate the HSI rotation problem. Zheng et al. [41] constructed two novel attention modules to avoid the impact of the interfering pixels and extract rotation-invariant spectral–spatial features. Although substantial CNN-based classification methods have achieved satisfactory performance, they still have some defects as follows.

1) CNNs expand the receptive field by stacking convolutional layers and pooling layers, but the actual receptive field is still insufficient, so it has some limitations in capturing global feature representation.
2) The extraction of spectral features in HSI is usually realized by 3-D convolution, so the parameters of the model will be large.

Transformer [42] was first proposed to be used in the natural language processing (NLP) field in 2017 and showed its great potential. The transformer captures the global dependencies of sequence data through a multihead self-attention (MHSA) mechanism. In recent years, the transformer has been introduced into the field of CV [43], [44]. But these methods are used in natural image analysis. Some researchers have proposed transformer-based HSI classification methods. For example, Hong et al. [45] proposed SpectralFormer, which can learn spectral local sequence information from adjacent bands of HSI, thus generating grouped spectral embedding. Sun et al. [46] presented a spectral–spatial feature tokenization transformer (SSFTT) model to capture spectral–spatial features and high-level contextual features. However, there are some problems in the existing transformer-based methods, which limit the continuous development of transformers in HSI classification. The problems are summarized as follows.

1) The transformer captures the context relationship between all the tokens through MHSA. When the number of tokens is large, the computation will be heavy.
2) All transformer-based methods do not consider improving the discriminability of the extracted spectral and spatial features, nor do they consider the impact of both on the classification results.

To deal with the above-mentioned problems, in this article, an efficient lightweight spectral–spatial transformer (ELS2T) is proposed for HSI classification. In ELS2T, we design a separable spatial–spectral self-attention (S^3A) module to replace the MHSA module. Therefore, the calculation efficiency of the model has been significantly improved, and S^3A has spectral awareness. Besides, we design a global multiscale attention module (GMAM) to improve the discriminant power of the extracted features, and an adaptive feature fusion module (AFFM) is proposed to adaptively fuse the acquired spectral and spatial information. In general, the main contributions of this article are as follows.

1) We propose a novel HSI classification network called ELS2T, which is a lightweight transformer model.
2) We design a GMAM to highlight important information and suppress unimportant information, and propose an AFFM to obtain the importance of spectral and spatial features for classification tasks.
3) We design an S^3A module to replace the MHSA module to improve computational efficiency, and S^3A has spectral awareness. In addition, we introduced the gating mechanism to form the gated feed-forward network (GFN) so that useful information can be further disseminated.
4) Experimental results on four benchmark datasets show that the proposed ELS2T outperforms the other advanced DL methods in both classification performance and computational efficiency.

The remainder of the article is organized as follows. Section II provides related work on DL-based HSI classification and the application of transformer in CV. In Section III, the proposed ELS2T is introduced. Section IV describes the four HSI datasets, experimental settings, and experimental results. Summarize is provided in Section V.

II. Related Work

A. Attention Mechanism

The attention mechanism originates from the study of human vision. Humans have always selectively focused on one part of all information and ignored others. The attention mechanism in DL is similar to the human visual system, the core goal is to ignore irrelevant information and focus on key information [47], [48]. In recent years, the attention model is widely used in various fields, including human body recognition, image processing, and speech recognition, etc., and is one of the most noteworthy and in-depth core technologies in DL.

In CV tasks, the self-attention mechanism is often used, which can be denoted as follows:

$$\text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \omega\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d_K}}\right)\boldsymbol{V} \tag{1}$$

where ω represents the softmax function, $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ are the query, key, and value, respectively, and d_K is the dimension of $\boldsymbol{K}$. Self-attention mechanism obtains the attention weight map by calculating the similarity between each token vector and normalizing results by the softmax function.In addition to the self-attention mechanism, there are also channel and spatial attention modules represented by the "squeeze-and-excitation" (SE) module [47] and convolutional block attention

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module (CBAM) [48] in the CV domain. These attention modules explicitly model the interdependence between channels or spatial positions and automatically acquire the weights of all channels or spatial positions through learning. More discriminative features can be obtained through the attention mechanism, so researchers have embedded attention modules in the HSI classification model. Guo et al. [49] proposed a spectral–spatial connection attention mechanism for HSI classification. Li et al. [50] designed two attention modules to reduce the impact of redundant frequency bands and interference pixels.

However, when there are massive pixels in the image, the calculation amount of the self-attention module is huge. In this article, we design an S^3A module to effectively reduce the amount of computation, and the GMAM and the AFFM are proposed based on the attention mechanism.

B. Transformer in CV

In the field of CV, CNN has always occupied the mainstream position. However, with the successful application of transformers in the NLP field, more and more researchers are attracted to the use of transformers for CV tasks. Dosovitskiy et al. [43] proposed the vision transformer (ViT) model, which is the first attempt to apply a transformer to natural image classification. ViT first divides the image into patches with embedded positions to build a series of tokens and then inputs them into the transformer encoder to model the global long-range dependencies. However, ViT ignores local features, which reduces the distinguish ability between background and foreground. To overcome this problem, the researchers proposed some methods to combine the transformer and CNN. For example, Wu et al. [51] applied convolution to ViT to improve spatial information. Based on this, a novel model convolutional vision transformer (CvT) is proposed. Guo et al. [52] designed a convolutional neural networks meet vision transformers (CMT) with the ability of long-range modeling and CNN local feature extraction. Peng et al. [53] proposed a hybrid network structure, called Conformer, which integrates local feature representation and global feature representation under different resolutions in an interactive manner by means of a feature coupling unit (FCU). Chen et al. [54] proposed a new parallelization of MobileNet and transformer. However, all the above models are designed for downstream tasks related to natural images, including image classification, object detection, instance segmentation, etc. Very recently, inspired by transformer-based natural image classification models, although there are some transformer-based HSI classification methods [55], [56], [57], [58], they ignore the huge amount of computation of the transformer.

III. Proposed Method

In this section, we will introduce the proposed ELS2T. The overall structure of the ELS2T is shown in Fig. 1, It can be seen that ELS2T consists of a spectral–spatial feature extraction network, a GMAM, an AFFM, and a lightweight transformer encoder (LTE). Specifically, the input of the ELS2T is fixed-size HSI cube, which is fed into the spectral and spatial CNN to extract shallow spatial–spectral features. These features are further fed into GMAM to improve their representation capabilities. Then, the enhanced spectral–spatial features are fed into AFFM to obtain their importance. Subsequently, shallow features are flattened and fed into LTE to capture semantic features. Finally, a multilayer perceptron (MLP) network is used to obtain the classification result.

A. Spectral–Spatial Feature Extraction

Suppose $\boldsymbol{X} \in \mathbf{R}^{h\times w\times c}$ is the original HSI data, where h and w are the width and height of the HSI cube, and c is the number of spectral channels. Therefore, HSI consists of c bands, providing useful spectral information, but there is redundancy. Therefore, we use the principal component analysis (PCA) algorithm to reduce the number of channels from c to b. Hence, HSI data processed by the PCA algorithm can be expressed as $\boldsymbol{X}_{\text{pca}} \in \mathbf{R}^{h\times w\times b}$. Next, we extract patches with each pixel of $\boldsymbol{X}_{\text{pca}}$ as the center, where the real label of each patch is the label of the center pixel. Then, we fed patches into a spectral CNN and a spatial CNN to extract local spectral information and local spatial information, respectively. The architecture of the spectral CNN and spatial CNN are illustrated in Fig. 2.

For the spectral CNN, we first use a strided 3-D convolution with a kernel size of $1 \times 1 \times 7$ to effectively reduce the spectral dimension. Then, the obtained feature maps are sent into three cascaded residual blocks, each of which contains two convolution layers with a kernel size of $1 \times 1 \times 7$, each convolution layer followed by a batch normalization (BN) layer and a rectified linear unit (ReLU) activation function. Finally, the output feature maps pass another 3-D convolution with a kernel size of $1 \times 1 \times (b-7)/2+1$. For the spatial CNN, the structure is similar to spectral CNN, but the kernel size of the first 3-D convolution layer is $1 \times 1 \times b$. And then, like spectral CNN, it is also three cascading residual blocks, but the kernel size of the convolution layer is $3 \times 3 \times 1$.

B. Global Multiscale Attention Module

To improve the representation ability of spectral and spatial features, a GMAM is designed. GMAM obtains global features $\boldsymbol{F}_G$ by adding spectral features $\boldsymbol{F}_{\text{spe}}$ and spatial features $\boldsymbol{F}_{\text{spa}}$. By multiplying the attention weight, $\boldsymbol{F}_G$ can be used to capture channel correlation and positional information, so as to effectively highlight the useful information and weaken the useless information. To generate attention weight, we have designed the multiscale coordinate attention module (MCAM), which replaces the global pooling used in the traditional channel attention module with two kinds of strip pooling operations, so the location information is embedded. In addition, to improve the sensitivity to key information, we also introduce multiscale information. Finally, through GMAM processing, more discriminative spectral features and spatial features are obtained. We describe the detailed operations below.

As shown in Fig. 3, first, we fuse spectral features $\boldsymbol{F}_{\text{spe}}$ and spatial features $\boldsymbol{F}_{\text{spa}}$ to obtain global features $\boldsymbol{F}_G$. Therefore, $\boldsymbol{F}_G$ can be expressed as

$$\boldsymbol{F}_G = \boldsymbol{F}_{\text{spe}} + \boldsymbol{F}_{\text{spa}}. \tag{2}$$

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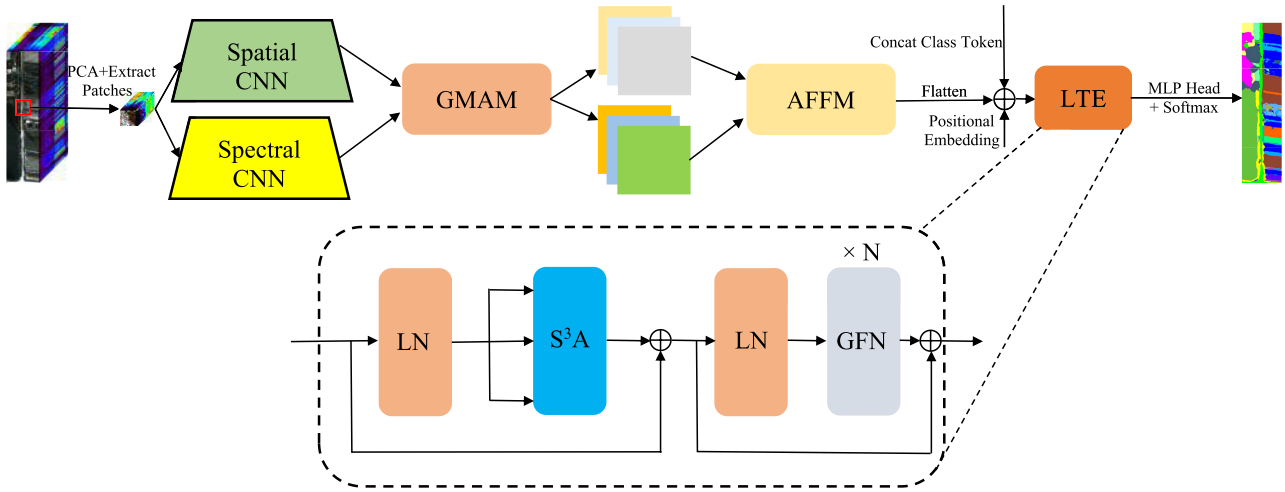


Fig. 1. Architecture of the proposed ELS2T for HSI classification. ⊕ Denotes element-wise summation and LN represents layer normalization.

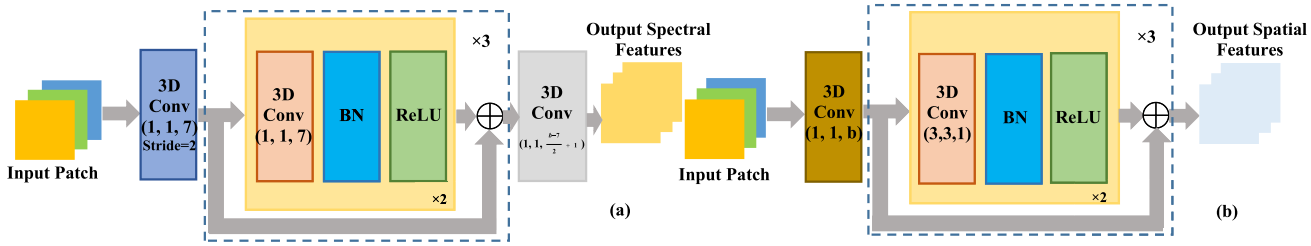


Fig. 2. Architecture of the spectral and spatial CNN. ⊕ Denotes element-wise summation. (a) Spectral CNN. (b) Spatial CNN.

We use two convolution operations to divide $\boldsymbol{F}_G$ into two parts, named $\boldsymbol{G}_{\text{spe}}$ and $\boldsymbol{G}_{\text{spa}}$, respectively. which can be summarized as

$$\boldsymbol{G}_{\text{spe}} = f^{3\times3}(\boldsymbol{F}_G) \tag{3}$$

$$\boldsymbol{G}_{\text{spa}} = f^{3\times3}(\boldsymbol{F}_G) \tag{4}$$

where $f^{3\times3}$ represents the convolution operation with a kernel size of 3×3. Then, we send $\boldsymbol{G}_{\text{spe}}$ and $\boldsymbol{G}_{\text{spa}}$ to an MCAM to obtain attention weight maps, generate multiscale information, and multiply the multiscale feature maps with the attention weights. Next, we will take $\boldsymbol{G}_{\text{spe}}$ as the input feature map as an example to introduce MCAM in detail.

It can be seen from Fig. 3 that MCAM contains two branches. The first branch is used to generate attention weights. Both average pooling and max pooling collect important clues about distinctive target features in HSI. Therefore, we use both average pooling and max pooling simultaneously. For a given feature map $\boldsymbol{G}_{\text{spe}}$, using horizontal average pooling, horizontal max pooling, vertical average pooling, and vertical max pooling to aggregate information along two spatial directions, we obtained a horizontal average pool feature z_{ha}, a horizontal max pool feature z_{hm}, a vertical average pool feature z_{va}, and a vertical max pool feature z_{vm}. Specifically, the calculation formula for each element in the horizontal direction is as follows:

$$z^c_{\text{ha}i} = \frac{1}{W}\sum_{j=0}^{W-1} z^c(i,j) \tag{5}$$

$$z^c_{\text{hm}i} = \max(z^c(i,j)), \quad j \in [0, W-1]. \tag{6}$$

The calculation formula for each element in the vertical direction is as follows:

$$z^c_{\text{va}j} = \frac{1}{H}\sum_{i=0}^{H-1} z^c(i,j) \tag{7}$$

$$z^c_{\text{hm}j} = \max(z^c(i,j)), \quad i \in [0, H-1]. \tag{8}$$

We add z_{ha} and z_{hm} to get z_h, z_{va} and z_{vm} to get z_v. Subsequently, we concatenate them along a spatial dimension, and send the concatenated feature maps into a shared 1×1 convolution layer, this process can be expressed as

$$\boldsymbol{M} = \delta(f^{1\times1}([z_{\text{ha}} + z_{\text{hm}}, z_{\text{va}} + z_{\text{vm}}])) \tag{9}$$

where $[\cdot,\cdot]$ denotes the concatenation operation, δ is the ReLU activation function, $f^{1\times1}$ denotes the 1×1 convolution, and $\boldsymbol{M}$ is an intermediate feature map encoding spatial information in two different directions. We then split $\boldsymbol{M}$ along the spatial dimension into two separate feature maps $\boldsymbol{M}_h$ and $\boldsymbol{M}_w$. To have the same number of channels as $\boldsymbol{G}_{\text{spe}}$, we use the other two convolution operations to apply to $\boldsymbol{M}_h$ and $\boldsymbol{M}_w$, respectively. Then two attention weights $\boldsymbol{F}_h$ and $\boldsymbol{F}_w$ are obtained after sigmoid function normalization. This process can be denoted as

$$\boldsymbol{F}_h = \sigma(f^{1\times1}(\boldsymbol{M}_h)) \tag{10}$$

$$\boldsymbol{F}_w = \sigma(f^{1\times1}(\boldsymbol{M}_w)) \tag{11}$$

where σ is the sigmoid function, and $f^{1\times1}$ denotes the 1×1 convolution operation. Another branch of MCAM uses three different dilate convolutions to generate multiscale features $\boldsymbol{G}^{\text{ms}}_{\text{spe}}$. The output features of the MCAM $\boldsymbol{G}'_{\text{spe}}$ are obtained

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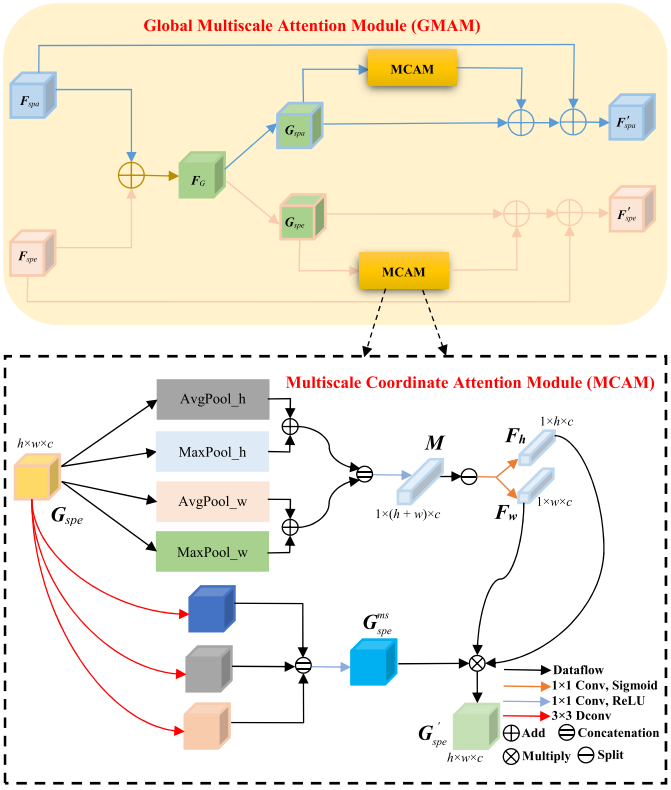


Fig. 3. Illustration of the GMAM. The GMAM contains two branches. A branch is used to generate attention weights, another branch uses three different dilate convolutions to generate multiscale features. AvgPool_h, MaxPool_h, AvgPool_w, and MaxPool_w are horizontal average pooling, horizontal max pooling, vertical average pooling, and vertical max pooling, respectively.

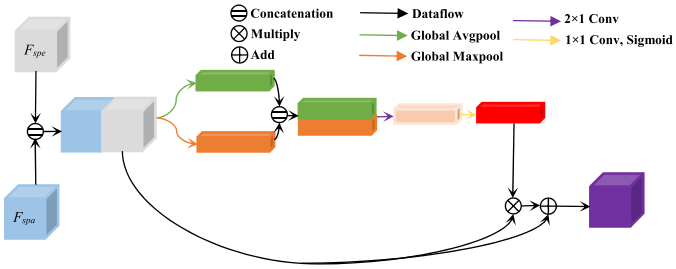


Fig. 4. Illustration of the AFFM.

by multiplying $\boldsymbol{F}_h, \boldsymbol{F}_w$, and $\boldsymbol{G}_{\text{spe}}^{\text{ms}}$, which can be expressed as

$$\boldsymbol{G}'_{\text{spe}} = \boldsymbol{G}_{\text{spe}}^{\text{ms}} \times \boldsymbol{F}_h \times \boldsymbol{F}_w. \tag{12}$$

Finally, to prevent overfitting, we apply residual connection to add $\boldsymbol{G}'_{\text{spe}}$, $\boldsymbol{G}_{\text{spe}}$, and $\boldsymbol{F}_{\text{spe}}$ to generate the enhanced spectral features $\boldsymbol{F}'_{\text{spe}}$. Similarly, the enhanced spatial features $\boldsymbol{F}'_{\text{spe}}$ can be obtained with the same operation.

C. Adaptive Feature Fusion Module

The importance of spectral features and spatial features for classification is different. To obtain their importance, we designed an AFFM to make the model more suitable for the HSI classification task. The structure of AFFM is illustrated in Fig. 4.

For given spectral features $\boldsymbol{F}_{\text{spe}}$ and given spatial features $\boldsymbol{F}_{\text{spa}}$, we first concatenate them to obtain $\boldsymbol{F}_f$, then we use global average pooling and global max pooling to obtain two channel descriptor and use a 2×1 convolution operation to fuse the two descriptors. Next, we apply 1×1 convolution to capture the relationship between channels and obtain weight coefficients through the sigmoid activation function, then use weight coefficients to weigh the concatenated features. Finally, to avoid gradient disappearance, we introduce the residual connection to add the weighted feature maps and the original concatenated feature maps.

Therefore, AFFM can be expressed as

$$\begin{aligned} \boldsymbol{F}_f &= [\boldsymbol{F}_{\text{spe}}; \boldsymbol{F}_{\text{spa}}] \\ \boldsymbol{M}(\boldsymbol{F}_f) &= \sigma(f^{1\times1}(f^{2\times1}([\text{MaxPool}(\boldsymbol{F}_f); \text{AvgPool}(\boldsymbol{F}_f)]))) \\ \boldsymbol{F} &= \boldsymbol{M}(\boldsymbol{F}_f) \times \boldsymbol{F}_f + \boldsymbol{F}_f \end{aligned} \tag{13}$$

where $f^{1\times1}$ denotes the 1×1 convolution, $f^{2\times1}$ represents the 2×1 convolution, $\boldsymbol{M}(\boldsymbol{F}_f)$ is the weight vector, and $\boldsymbol{F}$ is the output of AFFM.

D. Lightweight Transformer Encoder

To adapt to the input requirements of LTE, the output feature maps of AFFM are flattened into a token sequence $\boldsymbol{T}_f \in \mathbf{R}^{n\times d}$. Then, a learnable classification token $\boldsymbol{t}_{\text{cls}} \in \mathbf{R}^d$ is added at the beginning of the sequence and used to complete the final classification task. To retain location information, a position embedding $\boldsymbol{PE}_{\text{pos}} \in \mathbf{R}^{(n+1)\times d}$ is added to each token. Specifically, we adopt absolute position encoding, which uses a value to represent the absolute position of each token and is based on a learnable approach. The input $\boldsymbol{T}_0$ of LTE can be expressed as

$$\boldsymbol{T}_0 = [\boldsymbol{t}_{\text{cls}}; \boldsymbol{T}_f] + \boldsymbol{PE}_{\text{pos}}. \tag{14}$$

The MHSA module in ViT's TE uses the dot product to capture the context relationships between all tokens. However, when the number of tokens is large, the amount of computation is huge. Second, the MHSA module does not have spectral awareness, which limits its development in HSI classification. Therefore, we design the S^{3}A module to relieve the above problems. In addition, to further improve the representation learning ability, we introduced the gating mechanism to the feed-forward network (FFN) to form the GFN. LTE consists of several lightweight transformer blocks. Each block contains an S^{3}A layer and a GFN layer, which can be formulated as

$$\begin{aligned} \boldsymbol{T}'_i &= \text{S}^3\text{A}(\text{LN}(\boldsymbol{T}_{i-1})) + \boldsymbol{T}_{i-1}, \quad i = 1, \ldots, L \\ \boldsymbol{T}_i &= \text{GFN}(\text{LN}(\boldsymbol{T}'_i)) + \boldsymbol{T}'_i, \quad i = 1, \ldots, L \end{aligned} \tag{15}$$

where $\boldsymbol{T}_{i-1}$ denotes the token sequence output from block $i-1$, $\boldsymbol{T}'_i$ represents the sequence generated by the S^{3}A module, $\boldsymbol{T}_i$ is the output token sequence of block i, and LN represents layer normalization. Next, we will describe S^{3}A and GFN in detail.

1) Separable Spatial–Spectral Self-Attention: As illustrated in Fig. 5, the input token sequence $\boldsymbol{X} \in \mathbf{R}^{n\times d}$ is processed using four branches, i.e., input I, key K, value V, and spectral S. Input branch I apply a linear layer with weight $\boldsymbol{W}_I \in \mathbf{R}^d$ to convert each token in the input sequence into a scalar, $\boldsymbol{W}_I$ is regarded as a potential token. Linear mapping is a dot product

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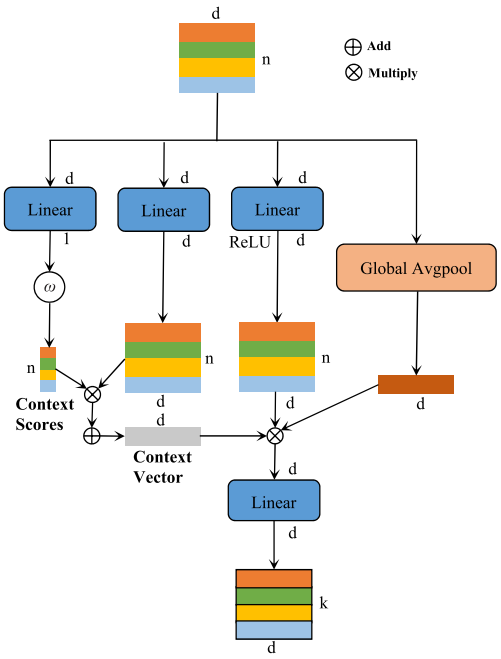


Fig. 5. Structure of the S^{3}A module. We note that ω is the softmax function.

operation, which calculates the similarity between the potential token and each token in $\boldsymbol{X}$, thus generating an n-dimensional vector. Then apply the softmax operation to the n-dimensional vector to generate the context score $\boldsymbol{c}_s \in \mathbf{R}^n$. Unlike the MHSA for calculating the attention score for each token relative to all tokens, the S^{3}A module only calculates the attention score for each token with respect to a potential token.

The context score $\boldsymbol{c}_s$ is used to calculate the context vector $\boldsymbol{c}_v$. Specifically, the input $\boldsymbol{X}$ is linearly projected using a linear layer with weight $\boldsymbol{W}_K \in \mathbf{R}^{d\times d}$ in the key branch to produce an output $\boldsymbol{K}$. The context vector $\boldsymbol{c}_v \in \mathbf{R}^d$ is calculated as the weighted sum of $\boldsymbol{K}$ as

$$\boldsymbol{c}_v = \sum_{i=1}^{k} \boldsymbol{c}_s(i)\boldsymbol{K}(i). \tag{16}$$

$\boldsymbol{c}_v$ encodes global information. So the input $\boldsymbol{X}$ is linearly projected using a linear layer with weight $\boldsymbol{W}_V \in \mathbf{R}^{d\times d}$ in the value branch, followed by a ReLU activation to produce an output $\boldsymbol{V}$. The global context information in $\boldsymbol{c}_v$ is then propagated to $\boldsymbol{V}$ through element-wise multiplication. In addition, to capture spectral–spatial correlation, we obtain spectral features $\boldsymbol{S}$ through a global pooling operation in the spectral branch S. The result is then fed to another linear layer with weight $\boldsymbol{W}_O \in \mathbf{R}^{d\times d}$ to produce the final output $\boldsymbol{Y} \in \mathbf{R}^{k\times d}$. The S^{3}A module can be defined as

$$Y = \left(\sum(\omega(\boldsymbol{X}\boldsymbol{W}_I) \times \boldsymbol{X}\boldsymbol{W}_K) \times \delta(\boldsymbol{X}\boldsymbol{W}_V) \times S\right)\boldsymbol{W}_O. \tag{17}$$

2) Gated Feed-Forward Network: Specifically, GFN has two parallel paths, both paths contain a linear transformation layer, as shown in Fig. 6, one of which is activated by the Gaussian error linerar unit (GELU) function. Then perform element-wise multiplication on the output of the two paths. Finally, apply another linear layer to obtain the final output.

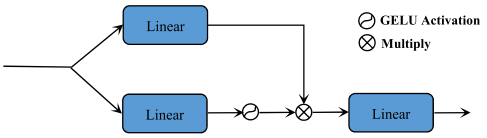


Fig. 6. Illustration of the GFN.

TABLE I
LAND-COVER CATEGORY AND NUMBERS OF TRAINING AND TEST SAMPLES ON THE IP DATASET

Class No.	Class Name	Training	Test
1	Alfalfa	2	44
2	Corn-notill	71	1357
3	Corn-mintill	42	788
4	Corn	12	225
5	Grass-pasture	24	459
6	Grass-trees	37	693
7	Grass-pasture-mowed	1	27
8	Hay-windrowed	24	454
9	Oats	1	19
10	Soybean-notil	49	923
11	Soybean-mintill	123	2332
12	Soybean-clean	30	563
13	Wheat	10	195
14	Woods	63	1202
15	Buildings-Grass-Trees-Drives	19	367
16	Stone-Steel-Towers	5	88
	Total	513	9736

Therefore, for a given input tensor $\boldsymbol{Z} \in \mathbf{R}^{k\times d}$, the process of GFN can be expressed as follows:

$$\boldsymbol{O} = (\phi(\boldsymbol{Z}\boldsymbol{W}^0) \times \boldsymbol{Z}\boldsymbol{W}^1)\boldsymbol{W}^2 \tag{18}$$

where $\boldsymbol{O}$ is the output of GFN and Φ represents GELU function. $\boldsymbol{W}^0 \in \mathbf{R}^{d\times 2d}$, $\boldsymbol{W}^1 \in \mathbf{R}^{d\times 2d}$, $\boldsymbol{W}^2 \in \mathbf{R}^{2d\times d}$ indicate weights of the three linear layers, respectively.

IV. EXPERIMENTS

In this section, we first describe the details of the four HSI datasets. Then, we introduce the experimental setup, including evaluation indicators, implementation details, and comparison methods. Finally, we compared the classification results and model complexity of all methods and conducted ablation experiments to verify the effectiveness of the proposed ELS2T.

A. Datasets Description

In this article, four publicly available HSI datasets are used to evaluate the effectiveness of the proposed method, including the Indian Pine dataset (IP), WHU-Hi-HanChuan dataset (WHU-HC), WHU-Hi-HongHu dataset (WHU-HH), and Botswana dataset (BS). Tables I–IV shows the land-cover category and numbers of training and test samples.

1) IP Dataset: This dataset is the earliest dataset for HSI classification and a piece of the IP tree in IN, USA, was imaged by the Airborne Visible Imaging Spectrometer (AVIRIS) in 1992. AVIRIS continuously images ground objects in 220 consecutive bands, but since the 104th–108th,

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TABLE II
LAND-COVER CATEGORY AND NUMBERS OF TRAINING AND TEST SAMPLES ON THE WHU-HC DATASET

Class No.	Class Name	Training	Test
1	Strawberry	45	44690
2	Cowpea	23	22730
3	Soybean	10	10277
4	Sorghum	5	5348
5	Water spinach	1	1199
6	Watermelon	5	4528
7	Greens	6	5897
8	Trees	18	17960
9	Grass	9	9460
10	Red roof	11	10505
11	Gray roof	17	16894
12	Plastic	4	3675
13	Bare soil	9	9107
14	Road	19	18541
15	Bright object	1	1135
16	Water	76	75325
	Total	259	257271

TABLE III
LAND-COVER CATEGORY AND NUMBERS OF TRAINING AND TEST SAMPLES ON THE WHU-HH DATASET

Class No.	Class Name	Training	Test
1	Red roof	14	14027
2	Road	4	3508
3	Bare soil	22	21799
4	Cotton	163	163122
5	Cotton firewood	6	6212
6	Rape	45	44512
7	Chinese cabbage	24	24079
8	Pakchoi	4	4050
9	Cabbage	11	10808
10	Tuber mustard	12	12382
11	Brassica parachinensis	11	11004
12	Brassica chinensis	9	8945
13	Small Brassica chinensis	22	22485
14	Lactuca sativa	7	7349
15	Celtuce	1	1001
16	Film covered lettuce	7	7255
17	Romaine lettuce	3	3007
18	Carrot	3	3214
19	White radish	9	8703
20	Garlic sprout	3	3483
21	Broad bean	1	1327
22	Tree	3	4037
	Total	384	386309

150th–163rd, and 220th bands cannot be reflected by water; therefore, we generally use the remaining 200 bands after removing these 20 bands. The HSI contains 145 × 145 pixels and 16 land-cover classes.

2) WHU-HC Dataset: The WHU-HC dataset was collected on June 17, 2016, in Hanchuan, Hubei, China, with a 17-mm focal length Headwall Nano-Hyperspec imaging sensor, powered by a Leica Aibot X6 UAV V1 platform. The study area is an agricultural scene in an urban–rural area. The spatial resolution is about 0.109 m and there are 274 bands with spectral wavelengths ranging from 400 to 1000 nm. The image contains 1217 × 303 pixels and 16 land-cover classes.

3) WHU-HH Dataset: The WHU-HH dataset was collected on November 20, 2017, in Honghu, Hubei, China, on a DJI Matrix equipped with a 17 mm focal length Headwall Nano-Hyperspec imaging sensor 600 Pro UAV platform. The experimental area is a complex agricultural scene with a wide variety of crops. The spatial resolution is about 0.043 m and there are 270 bands with spectral wavelengths ranging from 400 to 1000 nm. The image contains 940 × 475 pixels and 22 land-cover classes.

TABLE IV
LAND-COVER CATEGORY AND NUMBERS OF TRAINING AND TEST SAMPLES ON THE BS DATASET

Class No.	Class Name	Training	Test
1	Water	14	256
2	Hippo grass	5	96
3	Floodplain grasses1	13	238
4	Floodplain grasses2	11	204
5	Reeds1	13	256
6	Riparian	13	256
7	Firescar2	13	246
8	Island interior	10	193
9	Acacia woodlands	16	298
10	Acacia shrublands	12	236
11	Acacia grasslands	15	290
12	Short mopane	9	172
13	Mixed mopane	13	255
14	Exposed soils	5	90
	Total	162	3086

4) BS Dataset: The BS dataset is derived from a series of data obtained by NASAs EO-1 satellite in Botswana from 2001 to 2004. After removing the noise bands, the remaining 145 bands with spectral wavelengths ranging from 400 to 2500 nm were used for the experiments. The image contains 1476 × 256 pixels and 14 land-cover classes.

B. Experimental Setup

1) Evaluation Metrics: Three widely used evaluation metrics are selected to evaluate the classification performance of all methods, i.e., overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa). All the models have been run five times, and the average result and standard deviation are recorded.

2) Comparison Methods: To verify the effectiveness of the proposed algorithm, we selected several representative state-of-the-art HSI classification methods for comparison, including spectral–spatial residual network (SSRN) [34], HybridSN [31], enhanced multiscale feature fusion network (EMFFN) [35], adaptive multiscale feature attention network (AMFAN) [38], rotation-invariant attention network (RIAN) [41], SSFTT [46], and combine spatial attention and a multibranch prediction approach (SST-M) [57].

1) The SSRN is a CNN, which includes two spectral residual blocks and two spatial residual blocks to learn discriminative features. The size of convolution kernels is 7 × 1 × 1 and 1 × 3 × 3 in two kinds of block, respectively.
2) The HybridSN uses both 2-D and 3-D convolutions to extract features, which includes three 3-D convolution layers, one 2-D convolution layer, three full connection layers, and two dropout layers.

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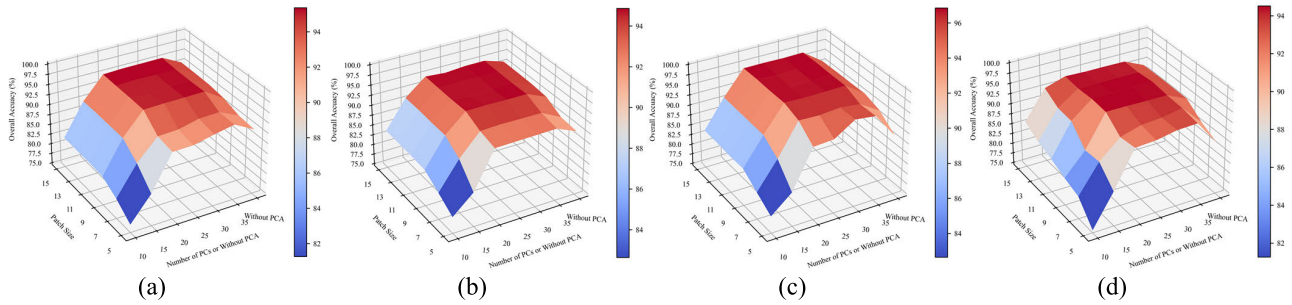


Fig. 7. Impact of PCA, number of PCs, and patch size on classification performance. (a) IP. (b) WHU-HC. (c) WHU-HH. (d) BS.

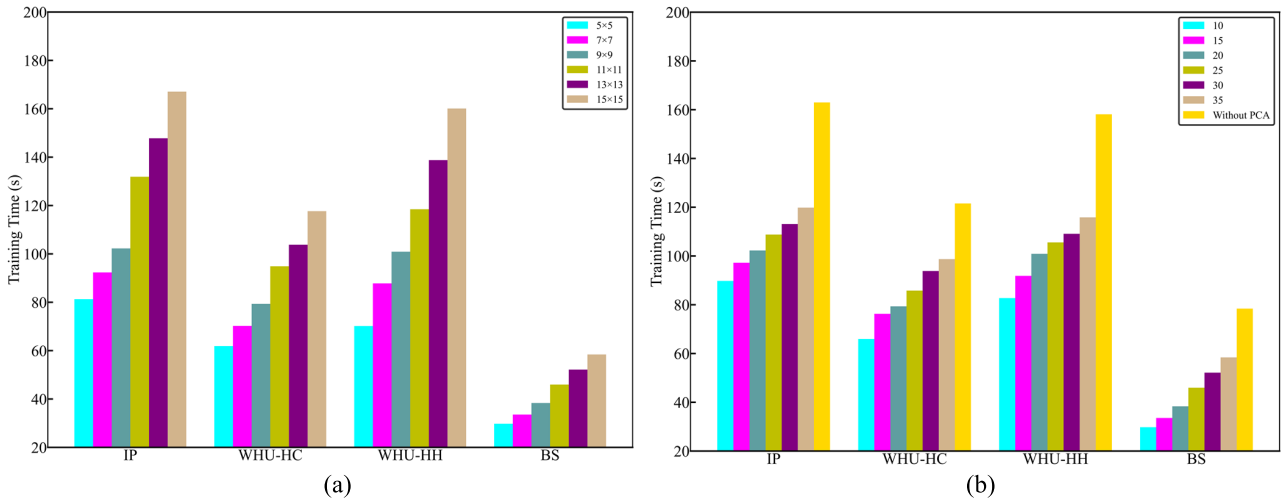


Fig. 8. (a) Impact of patch size on training time. (b) Impact of PCA and number of PCs on training time.

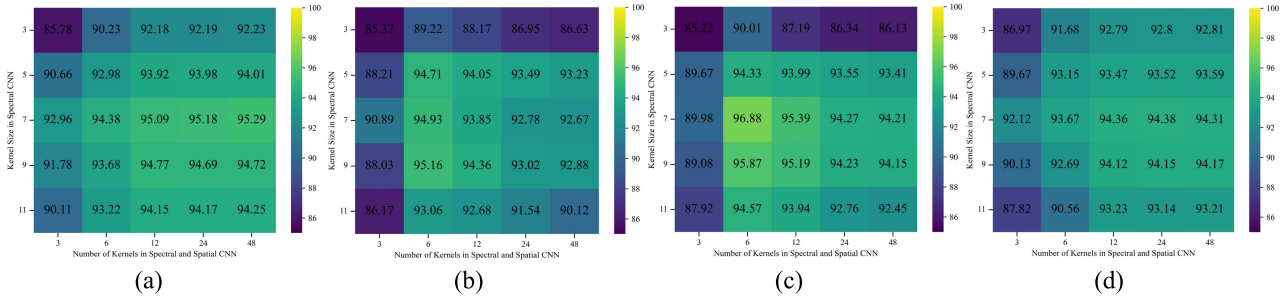


Fig. 9. Impact of kernel size in spectral CNN and the number of kernels in spectral CNN on OA. (a) IP. (b) WHU-HC. (c) WHU-HH. (d) BS.

3) For EMFFN, which consists of two networks named spectral cascaded dilated convolutional network (CDCN) and PMN. In the CDCN, four 6×1 dilated 2-D convolutional layers with different dilate rates are used to extract the spectral information. Besides, a channel attention module is applied to further capture the long-range information. In the PMN, reduce the dimensionality of the input cube through PCA. 2-D convolutional layers with 7×7, 5×5, and 3×3 kernels are applied to extract multiscale spatial features.
4) The AMFAN is a two-branch network. A branch is used for spectral feature extraction, where three cascaded 3-D residual blocks are used for spectral feature extraction of different levels, and a spectral attention module (SAM) is used to improve spectral feature discrimination. Another branch is used for spatial feature extraction. This branch obtains spatial features of different scales and calculates the importance of features of different scales through the adaptive multiscale feature residual module (AMFRM), and a position attention module (PAM) is used to improve spatial feature discrimination.
5) RIAN abandons using 3×3 convolutions to capture spatial information. The center spectral attention (CSpeA) module, the 1×1 convolution, and the rectified spatial attention (RSpaA) module are utilized to build the network.
6) The SSFTT is a transformer-based method. First, a 3-D convolution layer and a 2-D convolution layer are used to extract shallow spectral–spatial features. Then, a Gaussian-weighted feature tokenizer is applied

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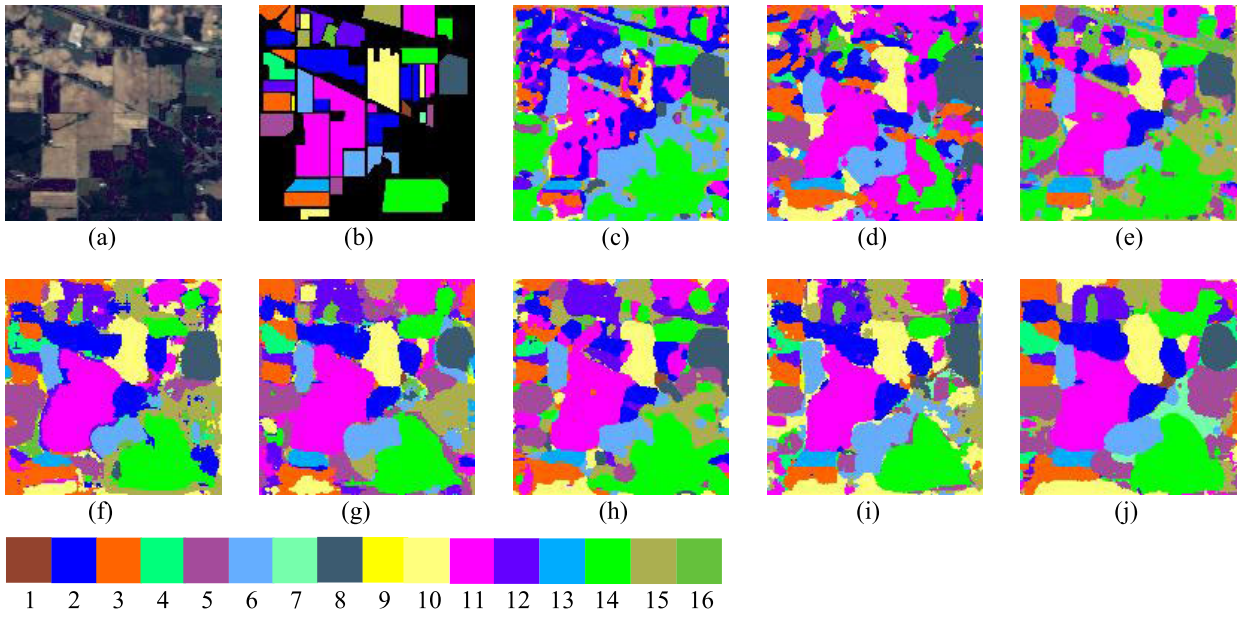


Fig. 10. Classification maps using different classification methods on the IP dataset. (a) False-color image. (b) Ground-truth map. (c) SSRN. (d) HybridSN. (e) EMFFN. (f) AMFAN. (g) RIAN. (h) SSFTT. (i) SST-M. (j) ELS2T.

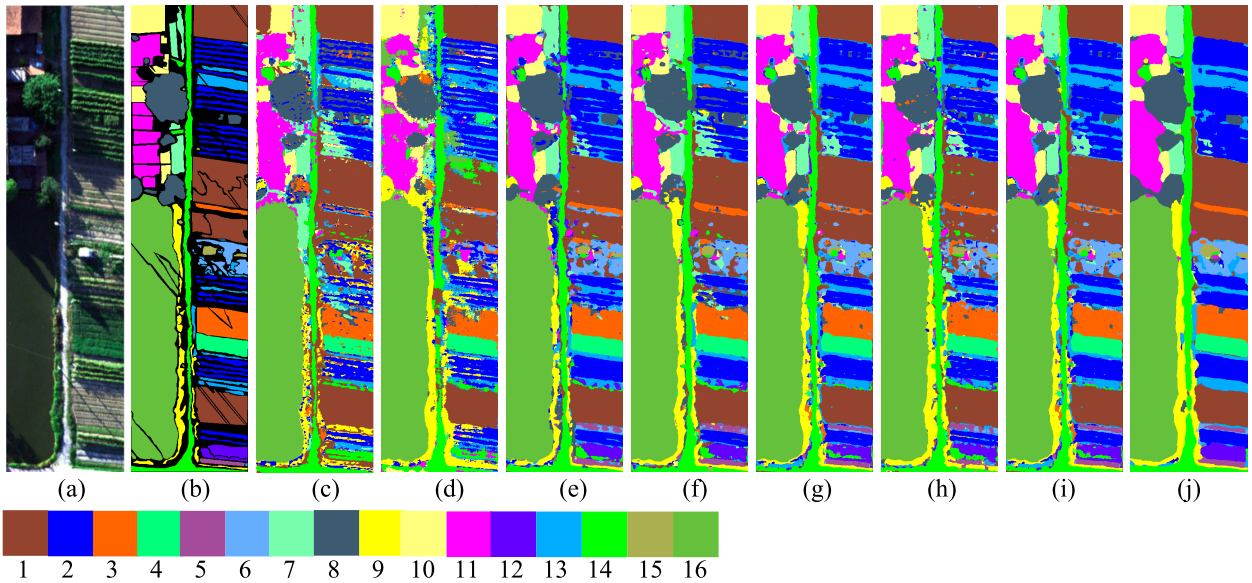


Fig. 11. Classification maps using different classification methods on the WHU-HC dataset. (a) False-color image. (b) Ground-truth map. (c) SSRN. (d) HybridSN. (e) EMFFN. (f) AMFAN. (g) RIAN. (h) SSFTT. (i) SST-M. (j) ELS2T.

to transform the spatial–spectral features into semantic tokens. Finally, the semantic tokens are sent to the transformer encoder for feature learning.

7) The SST-M is a transformer-based method. In SST-M, a spectral feature extraction network is used to extract spectral information. Then, the spectral cube is divided and flattened to a sequence, and fed into the transformer encoders to capture long-range dependencies. Besides, spatial sequence attention (SSA) is used between each transformer encoder.

To make a fair comparison, for the training of ELS2T and all comparison networks, we use the Adam optimizer, the batch size is set to 64. The learning rate is set to 0.001, and the epoch is set to 100. The rest of the parameters are consistent with that described in the original paper.

3) Implementation Details: Our proposed ELS2T and the compared methods are implemented on the Pytorch platform. The hardware used for training was an Intel[1] Xeon[1] Gold 6130 CPU and an Nvidia Tesla V100-PCIE GPU.

C. Parameter Analysis

1) Input Patch Size, PCA and Number of Principal Components: The input patch size and the number of principal components (PCs) retained in the PCA algorithm determine

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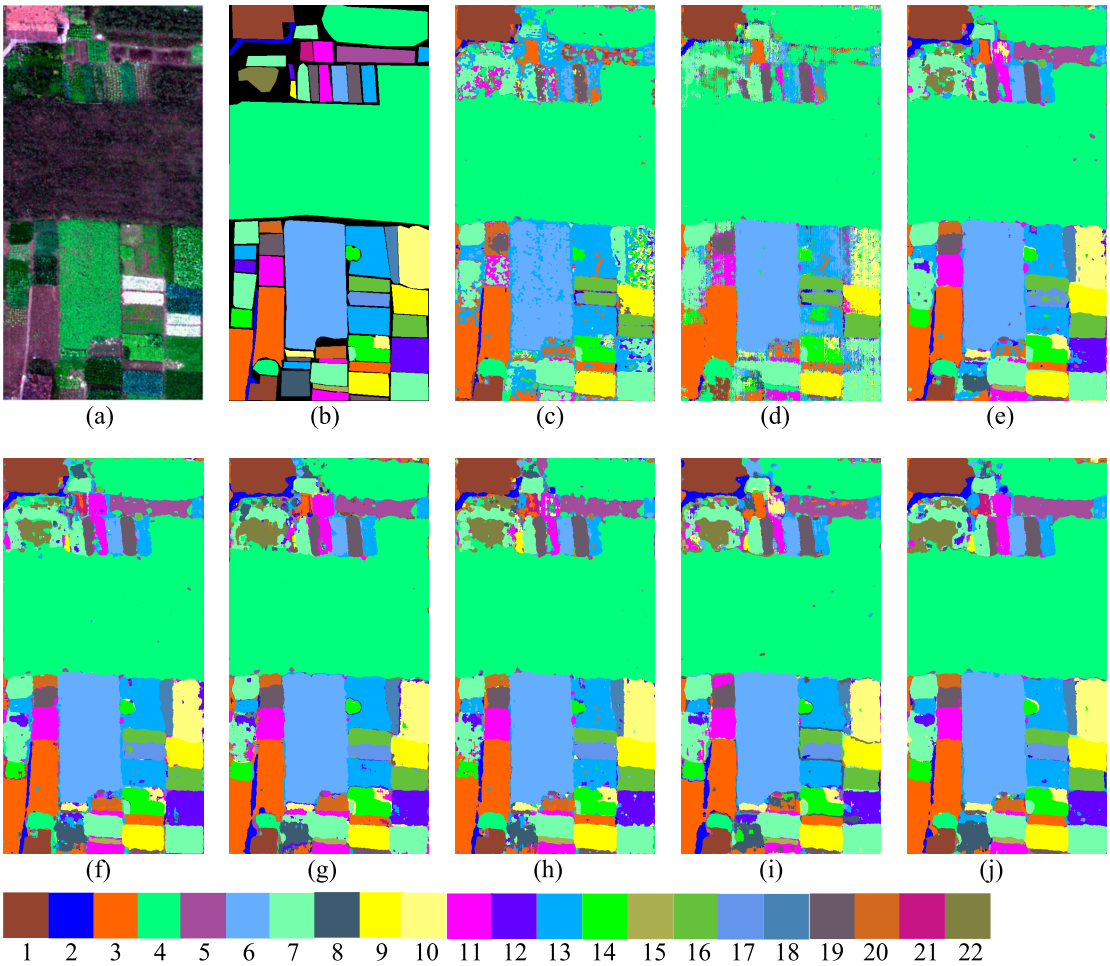


Fig. 12. Classification maps using different classification methods on the WHU-HH dataset. (a) False-color image. (b) Ground-truth map. (c) SSRN. (d) HybridSN. (e) EMFFN. (f) AMFAN. (g) RIAN. (h) SSFTT. (i) SST-M. (j) ELS2T.

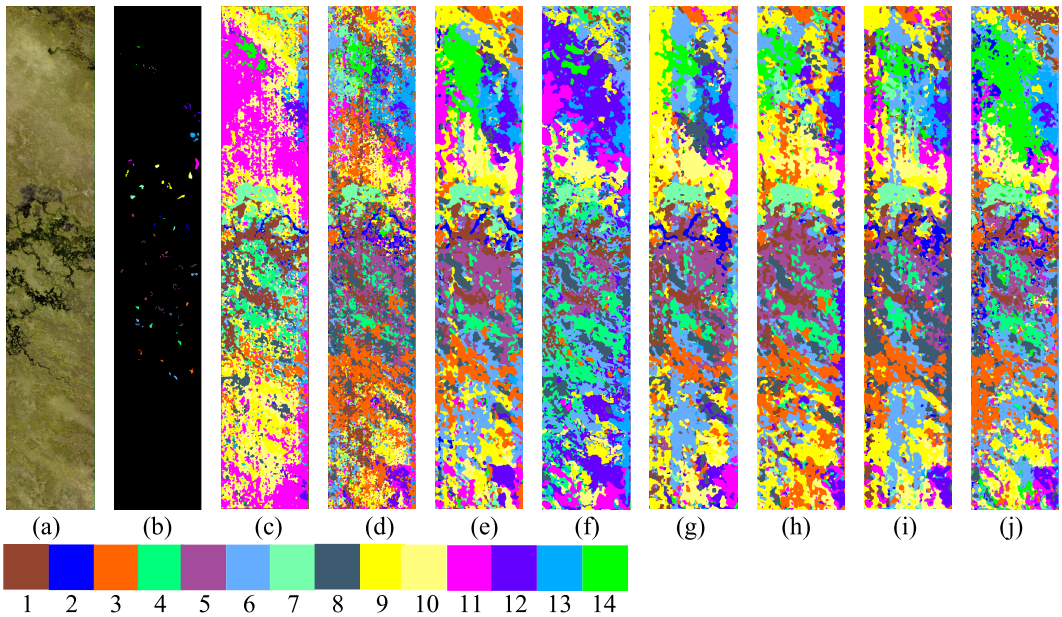


Fig. 13. Classification maps using different classification methods on the BS dataset. (a) False-color image. (b) Ground-truth map. (c) SSRN. (d) HybridSN. (e) EMFFN. (f) AMFAN. (g) RIAN. (h) SSFTT. (i) SST-M. (j) ELS2T.

the amount of information for each input, and have an important impact on whether ELS2T can work effectively. In this section, we vary the patch sizes from 5×5 to 15×15 and vary the number of PCs from 10 to 35 on the four datasets. In addition, to verify the effectiveness of PCA, we also attempted not to use PCA. Fig. 7 shows the impact of PCA, number of PCs, and patch size on the model performance. It can be observed that the patch size and the number of PCs

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TABLE V
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS FOR IP DATASET

Class	SSRN	HybridSN	EMFFN	AMFAN	RIAN	SSFTT	SST-M	ELS2T
1	17.73±2.03	11.82±2.03	32.27±3.80	45.45±0.00	38.64±0.00	56.09±0.00	43.18±0.00	**75.00±0.00**
2	60.96±6.98	51.51±5.86	55.84±3.80	89.14±1.52	84.92±2.69	57.11±0.40	86.59±0.68	**92.39±0.41**
3	49.64±4.73	72.74±9.16	78.81±4.13	93.88±1.05	87.46±2.02	82.74±1.21	94.90±1.05	**99.39±0.58**
4	18.40±1.35	19.11±1.26	12.18±0.49	80.00±2.27	92.18±2.85	52.44±0.40	75.29±1.01	**92.71±0.80**
5	33.33±2.96	54.42±4.43	75.95±6.62	93.25±1.74	94.07±1.88	80.61±1.00	95.03±1.52	**96.30±0.44**
6	92.06±11.61	85.05±10.65	92.41±7.40	86.44±1.24	93.56±2.30	**94.66±1.38**	94.17±1.11	93.16±1.36
7	22.96±3.31	12.59±4.06	11.85±3.31	38.52±4.06	62.22±9.66	29.63±3.31	59.26±0.00	**96.29±0.00**
8	93.48±10.73	92.56±10.74	95.46±6.73	96.21±2.05	96.30±3.29	97.36±0.62	96.52±1.34	**97.92±0.39**
9	20.00±4.71	17.89±5.77	24.21±5.77	63.16±0.00	31.58±0.00	31.58±0.00	**78.95±0.00**	57.89±0.00
10	28.86±4.41	67.37±8.99	70.21±5.59	86.31±1.29	89.10±2.45	78.22±0.98	92.13±1.26	**93.41±0.84**
11	74.92±8.42	77.62±8.78	89.86±6.40	89.45±1.59	95.67±2.65	92.20±0.89	94.10±1.10	**98.63±0.61**
12	39.18±5.11	34.71±4.53	37.12±2.25	80.28±0.83	82.06±2.62	82.59±0.73	73.93±0.58	**87.35±0.16**
13	67.79±7.45	77.23±8.10	95.28±7.99	76.72±1.86	91.49±1.56	81.54±0.86	96.62±0.56	**98.97±0.00**
14	91.74±11.96	54.23±6.17	93.07±6.21	96.89±1.66	97.42±2.86	**98.59±0.90**	97.25±1.30	96.82±0.25
15	45.50±4.27	19.07±1.02	75.10±5.50	80.44±1.41	81.41±2.68	89.37±0.71	**90.90±0.82**	87.08±0.99
16	31.59±1.02	35.91±3.05	87.73±6.31	69.55±1.02	78.18±2.49	21.59±0.00	74.77±1.01	**90.91±0.00**
OA(%)	63.24±7.35	62.88±7.24	76.63±5.40	88.75±1.49	90.78±2.53	82.43±0.83	91.28±1.03	**95.09±0.54**
AA(%)	49.26±5.45	49.01±5.74	64.21±4.35	79.10±1.41	81.02±2.51	69.97±0.78	83.97±0.79	**90.89±0.41**
Kappa	58.86±7.96	58.51±7.84	73.64±5.98	87.26±1.64	89.53±2.85	80.15±0.92	90.10±1.16	**94.42±0.61**

TABLE VI
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS FOR WHU-HC DATASET

Class	SSRN	HybridSN	EMFFN	AMFAN	RIAN	SSFTT	SST-M	ELS2T
1	96.24±2.18	89.43±2.05	95.75±4.06	96.96±2.54	96.69±5.01	97.27±2.60	98.13±1.95	**98.85±1.42**
2	80.75±2.28	84.94±7.35	92.62±5.13	92.65±0.79	93.31±1.90	90.63±2.39	93.19±1.65	**97.48±0.99**
3	75.58±5.02	74.24±3.25	91.89±2.29	94.57±4.14	89.44±2.17	95.52±2.62	97.10±2.06	**97.20±1.30**
4	85.77±4.72	94.48±3.90	96.16±5.02	97.29±2.99	96.01±3.24	97.77±2.85	96.72±2.03	**97.90±1.29**
5	17.00±1.51	5.74±0.34	78.18±4.81	91.36±6.63	86.48±5.72	73.58±2.14	87.13±1.70	**97.80±1.26**
6	8.91±0.09	27.57±0.95	73.75±8.43	72.77±1.82	80.21±5.27	68.63±1.91	72.62±1.48	**84.63±1.11**
7	89.07±7.65	27.07±1.10	86.12±3.67	**92.92±3.98**	90.48±5.37	90.39±2.62	88.67±1.91	90.76±1.24
8	70.04±4.17	63.62±2.45	85.42±10.47	85.33±1.93	87.77±4.18	88.73±2.44	94.11±1.39	**96.34±1.28**
9	46.04±1.64	81.78±3.83	66.96±1.42	91.37±2.10	84.47±2.64	89.28±2.41	91.90±2.01	**95.23±1.15**
10	67.89±4.05	89.49±2.79	95.90±2.56	92.81±2.32	**96.87±4.39**	98.34±2.97	97.24±1.83	95.66±1.15
11	87.81±8.26	79.31±3.42	94.53±3.24	95.33±2.18	95.95±2.58	96.54±2.85	96.40±1.92	**97.96**±1.25
12	12.88±0.28	11.63±1.55	73.57±2.94	78.72±2.89	76.76±1.68	70.07±2.25	79.87±1.96	**94.62**±1.20
13	39.89±1.55	56.96±5.73	70.06±4.12	62.90±6.08	76.76±2.85	65.32±2.02	77.84±1.60	**89.22**±1.24
14	83.88±2.39	79.99±5.42	89.21±4.36	90.92±8.22	90.47±4.68	92.73±2.95	94.34±2.09	**95.96**±1.29
15	18.64±2.02	11.01±1.31	62.70±8.38	71.05±7.21	48.15±2.73	74.73±2.76	71.01±1.77	**77.02**±0.58
16	94.62±6.04	97.49±2.08	94.92±6.81	97.18±2.97	98.28±2.74	**99.40±2.94**	98.26±2.26	99.00±1.30
OA(%)	68.52±3.69	72.57±3.10	88.64±4.79	90.35±3.06	90.87±3.51	88.84±2.60	91.27±1.90	**94.93±1.25**
AA(%)	60.94±3.14	60.92±2.81	84.23±4.67	87.76±3.39	86.76±3.42	86.02±2.54	89.66±1.85	**94.11±1.19**
Kappa	60.39±4.23	64.72±3.67	85.30±6.01	87.48±3.88	88.18±4.45	85.56±3.27	88.69±2.41	**93.40±1.61**

have a significant impact on the classification performance. As the two parameters increase, the classification accuracy first increases and then tends to stabilize. In addition, the classification performance when PCA is not used is not as good as when the number of PCs is between 20 and 35. For the best balance between classification accuracy and time consumption, the most suitable patch size and the number of PCs for all datasets are 9 × 9 and 20, respectively.

Furthermore, we recorded the impact of patch size, PCA, and number of PCs on training time on the four datasets in Fig. 8. As can be seen from Fig. 8(a), as the patch size increases, the training time gradually increases. Fig. 8(b) show that the training time does not significantly increase as the number of PCs increases. As expected, there is a significant increase in training time when PCA is not used.

2) Kernel Size in Spectral CNN and the Number of Kernels in Spectral and Spatial CNN: The size of the kernel in spectral CNN represents the size of the spectral group, and the number of kernels in spectral CNN and spatial CNN represents the number of output feature maps. These two parameters have a significant impact on classification accuracy. Fig. 9 shows the impact of these two parameters on OA on the four datasets.

When fixed the kernel size in spectral CNN, for the IP dataset and BS dataset, when the number of convolution kernels in spectral and spatial CNN increases from 3 to 12, the OA is greatly improved. However, when the number of

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TABLE VII
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS FOR WHU-HH DATASET

Class	SSRN	HybridSN	EMFFN	AMFAN	RIAN	SSFTT	SST-M	ELS2T
1	95.01±2.15	87.83±1.51	92.59±3.38	95.01±2.19	97.02±2.02	95.53±3.49	95.39±3.22	**97.86±1.46**
2	52.44±2.01	39.58±1.24	83.79±2.78	80.50±1.33	81.79±3.37	78.04±3.51	**83.98**±2.25	79.75±2.05
3	91.16±4.47	93.84±1.10	93.20±2.85	94.02±3.52	92.29±3.93	95.76±1.98	93.18±3.10	**96.46±1.66**
4	97.31±3.45	99.28±1.08	96.99±3.88	97.71±3.07	97.68±3.50	97.48±4.65	97.31±2.74	**99.08±1.09**
5	41.18±3.42	12.53±0.12	85.06±5.75	84.93±2.80	87.28±3.13	87.32±2.00	88.12±2.00	**95.92±2.33**
6	81.93±3.82	94.05±1.26	95.24±4.53	96.20±3.39	97.58±2.24	97.72±2.20	98.57±1.59	**97.99±2.09**
7	84.08±5.02	86.96±1.15	93.06±2.99	92.01±2.63	88.93±2.20	91.99±2.19	92.73±1.64	**96.19±1.34**
8	54.79±3.70	22.15±0.42	53.82±4.99	82.68±0.91	72.93±2.35	68.53±1.30	74.73±3.16	**94.84±1.63**
9	89.88±2.61	88.34±1.89	92.17±4.65	95.36±3.56	93.58±3.13	96.27±4.44	94.32±3.19	**98.60±0.66**
10	28.35±0.95	55.44±0.66	90.91±3.79	85.95±3.18	85.00±4.67	92.38±4.09	91.37±1.95	**96.24±1.22**
11	32.90±1.42	51.15±0.92	79.72±4.44	90.01±1.93	92.31±2.85	81.31±3.13	68.48±2.07	**92.85±1.77**
12	16.38±0.47	10.85±0.08	74.10±9.28	83.97±3.17	86.03±1.56	77.53±2.43	73.73±3.33	**90.56±1.69**
13	82.99±2.87	55.27±1.17	83.77±9.50	90.36±3.14	91.02±2.71	82.58±1.95	86.48±1.15	**93.14±1.90**
14	64.07±3.73	73.11±1.76	90.95±4.07	89.67±2.57	**97.13±1.86**	87.61±2.32	90.04±0.33	95.83±1.98
15	50.83±2.64	31.79±0.59	86.71±8.06	82.38±4.19	85.13±1.11	**91.31±3.04**	81.20±1.72	83.56±1.11
16	91.26±2.37	92.97±1.14	92.63±4.05	92.50±1.45	95.11±4.18	86.05±2.38	92.78±2.16	**96.27±1.19**
17	22.51±1.02	19.49±0.47	77.02±2.83	90.26±3.79	94.41±3.34	**97.23±2.34**	90.14±2.88	96.25±2.20
18	45.78±1.69	47.33±1.16	84.18±9.74	91.64±3.17	91.18±3.22	90.29±3.31	70.98±0.85	**93.35**±2.02
19	60.08±2.62	77.21±1.33	90.23±2.84	85.35±2.05	89.40±2.23	85.63±2.18	**90.26±2.20**	89.43±2.32
20	64.18±8.42	10.54±0.10	67.45±2.54	77.81±3.97	**89.67±3.01**	71.54±2.78	25.56±0.59	87.99±1.70
21	58.09±4.49	16.26±0.07	19.47±1.12	70.13±2.64	24.16±0.08	50.87±1.86	6.53±0.08	**86.13±1.95**
22	17.60±3.11	35.06±0.58	62.46±2.15	78.97±1.94	92.43±2.52	88.04±1.58	84.83±0.92	**97.08±1.46**
OA(%)	81.58±3.24	82.65±1.05	91.75±4.24	93.80±2.88	94.15±3.00	93.13±3.36	92.30±2.22	**96.88±1.43**
AA(%)	60.13±2.92	54.59±0.87	81.16±4.33	87.61±2.66	87.37±2.61	85.95±2.60	80.49±1.88	**93.43±1.64**
Kappa	76.83±3.92	78.05±1.28	89.65±5.24	92.20±3.58	92.65±3.71	91.36±4.14	90.32±2.74	**96.07±1.79**

TABLE VIII
CLASSIFICATION RESULTS OBTAINED BY DIFFERENT METHODS FOR BS DATASET

Class	SSRN	HybridSN	EMFFN	AMFAN	RIAN	SSFTT	SST-M	ELS2T
1	99.14±2.02	97.89±2.25	98.20±3.06	99.30±1.69	98.20±2.74	99.60±0.55	98.75±2.17	**99.06±2.04**
2	99.17±2.28	93.75±2.55	98.33±4.80	76.46±1.14	95.42±6.00	49.17±2.28	96.67±4.27	**100.00±0.00**
3	**96.30±2.87**	95.38±3.36	76.13±1.92	68.49±3.71	93.45±1.27	95.63±2.27	83.03±1.27	75.63±1.03
4	98.24±2.15	97.45±3.28	92.16±3.67	98.43±2.72	70.59±3.02	97.94±3.69	64.12±3.05	**98.92±1.75**
5	60.39±0.70	**96.33±1.18**	93.60±2.85	81.41±1.88	68.67±1.40	91.64±3.89	68.59±1.80	84.22±0.70
6	96.41±2.24	80.23±1.62	83.59±1.99	**97.42±4.12**	89.30±3.39	96.33±2.32	73.36±1.40	83.83±2.04
7	98.29±2.10	96.34±3.68	99.19±1.63	98.37±2.07	99.11±1.76	97.40±4.01	99.27±2.18	**99.43±0.93**
8	91.30±2.68	97.82±2.58	96.89±4.21	97.72±3.78	96.27±3.14	98.13±2.70	98.24±2.27	**99.48±1.04**
9	94.16±2.63	92.89±1.45	94.56±4.73	82.89±2.32	77.65±1.47	98.32±2.01	97.18±2.35	**99.80±0.37**
10	31.36±0.60	91.78±3.03	98.31±3.17	87.37±2.96	98.47±3.03	90.17±3.46	97.88±3.00	**99.07±1.10**
11	98.41±2.32	96.83±1.65	94.48±2.63	98.62±2.53	76.21±2.76	85.79±2.46	92.34±3.50	**99.45±0.79**
12	**98.49±1.76**	68.60±1.42	69.88±1.91	72.09±5.39	74.77±1.33	96.98±5.16	95.81±3.80	87.67±1.91
13	81.96±1.11	96.39±5.18	93.80±3.61	96.31±4.46	90.75±2.52	81.65±4.42	69.25±2.33	**99.29±0.86**
14	92.44±2.43	77.11±2.98	87.78±2.72	97.56±2.43	82.67±4.61	79.56±2.53	55.78±2.90	**100.00±0.00**
OA(%)	87.48±1.84	92.40±2.39	91.57±2.87	90.73±2.76	86.28±2.37	92.06±2.87	85.91±2.33	**94.36±1.01**
AA(%)	88.29±1.84	91.34±2.43	91.21±2.90	90.17±2.73	86.54±2.61	89.88±2.90	85.02±2.45	**94.70±0.95**
Kappa	86.45±1.98	91.77±2.59	90.87±3.10	89.96±2.99	85.15±2.56	91.40±3.11	84.75±2.52	**93.89±1.10**

convolution kernels in spectral and spatial CNN increases from 12 to 48, the improvement of OA is not obvious. For the WHU-HC dataset and WHU-HH dataset, when the number of convolution kernels in spectral and spatial CNN increases from 3 to 6, the OA increases significantly. However, when the number of convolution kernels in spectral and spatial CNN increases from 6 to 48, the OA decreases. Therefore, for the IP dataset and the BS dataset, the optimal number of convolution kernels of spectral and spatial CNN is 12. For the WHU-HC dataset and the WHU-HH dataset, the best number of convolution kernels in spectral and spatial CNN is six.

When fixed the number of convolution kernels in spectral and spatial CNN, for all datasets, when the kernel size in spectral CNN increases from 3 to 7, the OA improved a lot. However, when the number of convolution kernels in spectral and spatial CNN increases from 7 to 11, the OA decreases. Therefore, the optimal kernel size in spectral CNN is seven for all datasets.

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TABLE IX
MODEL COMPLEXITY COMPARISON

Method	Params(M)	Flops(MMac)	Training Time(s)	Test Time(s)
IP				
SSRN	0.36	95.01	203.45	3.17
HybridSN	5.12	248.15	176.89	1.71
EMFFN	0.26	113.23	289.05	1.24
AMFAN	1.04	22.44	155.79	1.93
RIAN	0.09	4.53	166.87	1.03
SSFTT	0.15	11.40	165.06	0.80
SST-M	0.04	6.39	249.32	0.68
ELS2T	0.03	5.63	102.26	0.55
WHU-HC				
SSRN	0.48	130.77	158.91	103.71
HybridSN	5.12	248.15	149.65	40.70
EMFFN	0.27	113.58	240.37	43.98
AMFAN	1.08	22.75	176.76	56.82
RIAN	0.13	4.95	178.23	35.98
SSFTT	0.15	11.40	128.94	15.13
SST-M	0.06	10.39	244.26	18.73
ELS2T	0.01	1.48	79.35	5.88
WHU-HH				
SSRN	0.47	128.84	195.36	153.82
HybridSN	5.12	248.15	235.12	21.94
EMFFN	0.27	113.56	298.47	48.90
AMFAN	1.08	22.74	190.33	74.14
RIAN	0.13	4.93	211.69	22.89
SSFTT	0.15	11.40	159.92	26.88
SST-M	0.06	10.14	273.80	20.35
ELS2T	0.01	1.48	100.89	7.90
BS				
SSRN	0.28	68.91	86.78	1.08
HybridSN	5.12	248.15	65.25	0.56
EMFFN	0.27	113.12	78.68	0.61
AMFAN	1.02	22.28	59.88	0.82
RIAN	0.07	4.23	68.11	0.47
SSFTT	0.15	11.40	73.92	0.23
SST-M	0.04	5.92	98.80	0.25
ELS2T	0.03	5.64	38.34	0.17

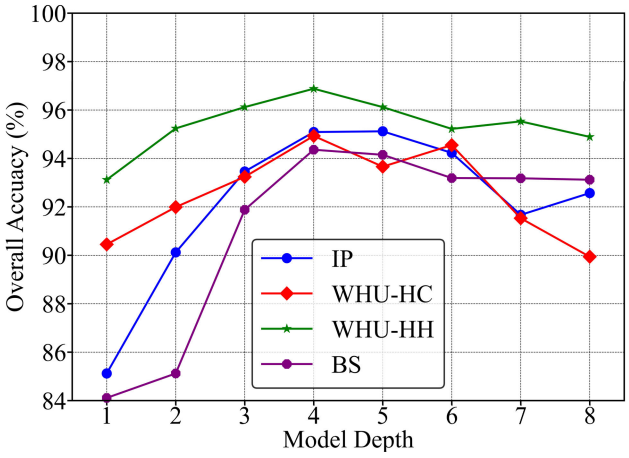


Fig. 14. Classification performance under different model depths.

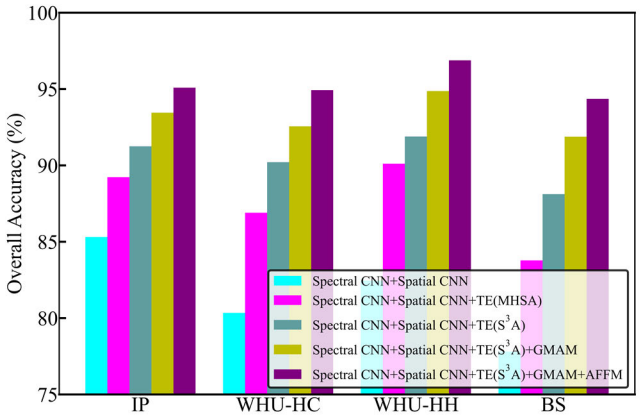


Fig. 15. Ablation study of the proposed modules.

D. Classification Results and Analysis

The classification maps obtained by all methods are illustrated in Figs. 10–13, and the classification accuracies of these methods are shown in Tables V–VIII.

1) IP Dataset: As shown in Table V, we only used 5% of the samples as the training set, so the comparison methods are difficult to obtain satisfactory classification results. The OA, AA, and Kappa of proposed ELS2T are superior to other comparison methods. SSRN and HybridSN have the lowest classification accuracy, especially for categories with small sample sizes, such as "alfalfa," "grass-pace-lowered," and "oats." This is because the expression ability of features extracted by these two methods is insufficient. SSFTT is a transformer-based method that effectively captures long-distance feature dependencies. Compared with the previous two methods, the classification accuracy of the category "alfalfa" has been greatly improved. Although AMFAN and RIAN are CNN-based methods, they introduce attention modules, so they also encode global information and achieve higher classification accuracy than SSFTT. However, the spatial and spectral information of the input HSI patches still cannot be constrained at the same time, and the importance of both cannot be obtained. In contrast, ELS2T has a strong discriminant feature extraction ability and obtains the importance of spectral and spatial features. In addition, it can effectively capture global context information, so it has higher classification accuracy. Furthermore, it can be seen from Fig. 10 that the classification maps of the proposed ELS2T have a smoother visual effect and less misclassification.

2) WHU-HC Dataset: As shown in Table VI, SSRN, and HybridSN obtain lower classification accuracy, indicating that they cannot effectively process small sample data and have weak expression ability. EMFFN carries out multiscale feature extraction and obtains the classification performance comparable to that attention model. In the transformer-based model, SST-M has achieved better classification performance than SSFTT. This may be mainly due to the role of the mask prediction branch in SST-M, which has been improved in the two categories "water spinach" and "bare soil." ELS2T has advantages in discriminant feature extraction and capturing long-range dependency, so the results show that the proposed ELS2T has a better classification effect. We can also see from Fig. 11 that ELS2T has obtained a cleaner classification map than the comparison methods.

3) WHU-HH Dataset: As shown in Table VII, SSRN, and HybridSN still have the worst classification accuracy, especially in the categories "celtuce" and "tree." It is worth noting that SSFTT has achieved better classification accuracy

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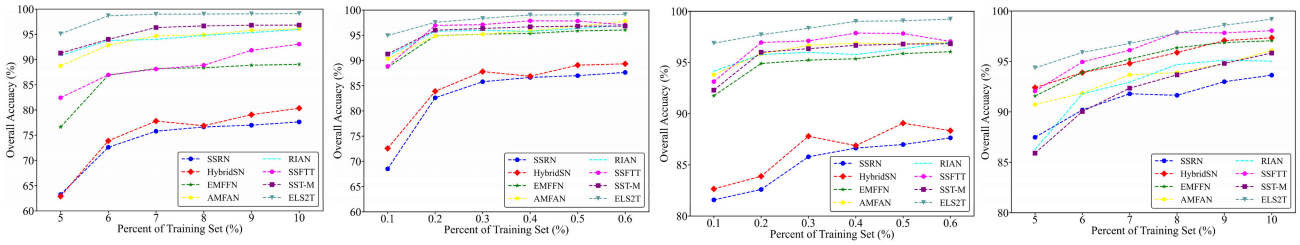


Fig. 16. Classification performance under different training samples. (a) IP. (b) WHU-HC. (c) WHU-HH. (d) BS.

than SST-M, which is different from the classification performance on the WHU-HH dataset. It can be seen from Fig. 11 that SSFTT has more advantages in large target recognition. RIAN has achieved a better classification effect than the transformer-based method, mainly because of its advantages in extracting rotation-invariant spectral–spatial features. The proposed ELS2T has been greatly improved in the categories "cotton firewood," "pakchoi," and "broad bean," and OA, AA, and Kappa are the best. We can also see from Fig. 12 that ELS2T has more correctly classified areas. This indicates that ELS2T can be better to extract discriminative features.

4) BS Dataset: As shown in Table VIII, compared to other datasets, HybridSN has a relatively high improvement in classification accuracy. SSFTT exhibits similar performance to HybridSN due to its ability to capture global information. ELS2T has been greatly improved in the categories "vineyard_untrained," and OA, AA, and Kappa are superior to other comparison methods. We can also see from Fig. 13 that ELS2T has fewer salt-and-pepper noises.

E. Model Complexity Analysis

In this section, to analyze the complexity of the proposed ELS2T and other comparison methods, we made statistics on the number of parameters, float operation per second (flops), average training and test time of five runs. For fairness, for all models, we use the Adam optimizer, and we set batch size, learning rate, and epoch to 64, $1e-3$, and 100, respectively. HybridSN, EMFFN, AMFAN, SSFTT, and ELS2T use PCA for dimensionality reduction, while SSRN, RIAN, and SST-M do not use PCA. The results are reported in Table IX. It can be seen that compared with other advanced DL methods, the proposed ELS2T has fewer parameters and flops on the premise of better classification accuracy, which is mainly due to the lightweight design of the S^3A module. For example, on the IP dataset, the flops of SSRN are close to 20 times that of ELS2T. It is worth noting that ELS2T has relatively high parameter quantities on the IP dataset and BS dataset, mainly because spectral CNN and spatial CNN use more convolution kernels on the IP dataset. In general, ELS2T can meet both high classification accuracy and inference speed.

F. Ablation Studies

1) Ablation Study on Model Depth: The model depth of ELS2T refers to the number of lightweight transformer blocks. Although the S^3A module effectively reduces the computational complexity, however, with the increase in the number of S^3A modules, the computational efficiency will still decline, and the classification accuracy will also be affected. To find the best balance between classification performance and calculation efficiency, we fixed all other parameters, and vary the model depth from 1 to 8 on the four datasets. Fig. 14 reports the OAs under different model depths on all datasets. It can be seen that the OA is the best when the model depth is four on all datasets.

2) Ablation Study of the Proposed Modules: This ablation analysis examines the effectiveness of all modules in the proposed ELS2T, including S^3A, GMAM, and AFFM. We conducted ablation experiments on the four datasets by adding different modules. The summary results are shown in Fig. 15. It is clear that when only spectral CNN and spatial CNN are used, the classification result is the worst on the four datasets. However, when TE using MHSA is introduced, the OA is significantly improved, which shows that global information is useful for HSI classification. After replacing the MHSA module with the S^3A module, OA is further improved, which proves that the S^3A module is not only lightweight but also improves the classification effect. When GMAM is added continuously, the classification performance is further improved. After AFFM is added, the classification accuracy reaches the highest.

3) Ablation Study of the Percentage of Training Samples: We studied the OAs of ELS2T and comparison methods under the different number of training samples. For the IP dataset and BS dataset, the percent of the training sample is both set to 5%–10%, and the percent of the training sample is both set to 0.1%–0.6% for the WHU-HC dataset and WHU-HH dataset. As shown in Fig. 16, on the four datasets, the proposed ELS2T achieves the best OA, and the OA increases monotonously with the increase of training samples. When there are few training samples, the ELS2T shows significant improvement compared with other comparison methods. When the number of training samples is large, ELS2T still has advantages in all methods, but the improvement is not particularly significant. In general, the OA of ELS2T is improved more significantly with limited samples.

V. Conclusion

In this article, we propose an ELS2T for HSI classification, which can model the long-range global dependencies in the HSI cube. First, a GMAM is designed to effectively highlight the useful information and weaken the useless information. Second, an AFFM is proposed to adaptively fuse the acquired spectral and spatial information. Finally, we design the S^3A

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module to replace the MHSA module in the TE to effectively reduce the complexity of the model and introduced the gating mechanism to the FFN to improve the representation learning ability. Experimental results on the four HSI datasets show that ELS2T is superior to other advanced DL methods in computing efficiency and classification performance. In future work, we will explore the semisupervised lightweight transformer model, further reduce the use of labeled HSI data and extract valuable features from unlabeled data.

REFERENCES

[1] G. ElMasry, D.-W. Sun, and P. Allen, "Non-destructive determination of water-holding capacity in fresh beef by using NIR hyperspectral imaging," *Food Res. Int.*, vol. 44, no. 9, pp. 2624–2633, Nov. 2011.

[2] Z. Xue, P. Du, J. Li, and H. Su, "Sparse graph regularization for robust crop mapping using hyperspectral remotely sensed imagery with very few in situ data," *ISPRS J. Photogramm. Remote Sens.*, vol. 124, pp. 1–15, Feb. 2017.

[3] K. Tan, F. Wu, Q. Du, P. Du, and Y. Chen, "A parallel Gaussian–Bernoulli restricted Boltzmann machine for mining area classification with hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 2, pp. 627–636, Feb. 2019.

[4] Y. Zhong, Q. Cao, J. Zhao, A. Ma, B. Zhao, and L. Zhang, "Optimal decision fusion for urban land-use/land-cover classification based on adaptive differential evolution using hyperspectral and LiDAR data," *Remote Sens.*, vol. 9, no. 8, p. 868, Aug. 2017.

[5] Y. Han, Y. Gao, Y. Zhang, J. Wang, and S. Yang, "Hyperspectral sea ice image classification based on the spectral–spatial-joint feature with deep learning," *Remote Sens.*, vol. 11, no. 18, p. 2170, Sep. 2019.

[6] S. Veraverbeke et al., "Hyperspectral remote sensing of fire: State-of-the-art and future perspectives," *Remote Sens. Environ.*, vol. 216, pp. 105–121, Oct. 2018.

[7] L. Ma, M. M. Crawford, and J. Tian, "Local manifold learning-based k-nearest-neighbor for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 11, pp. 4099–4109, Nov. 2010.

[8] J. Ham, Y. Chen, M. M. Crawford, and J. Ghosh, "Investigation of the random forest framework for classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 492–501, Mar. 2005.

[9] L. Bruzzone, M. Chi, and M. Marconcini, "A novel transductive SVM for semisupervised classification of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 11, pp. 3363–3373, Nov. 2006.

[10] G. Fan, M. Gan, B. Fan, and C. L. P. Chen, "Multiscale cross-connected dehazing network with scene depth fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jul. 1, 2022, doi: 10.1109/TNNLS.2022.3184164.

[11] G. Fan, Z. Hua, and J. Li, "Multi-scale depth information fusion network for image dehazing," *Appl. Intell.*, vol. 51, pp. 7262–7280, Mar. 2021.

[12] G.-D. Fan, B. Fan, M. Gan, G.-Y. Chen, and C. L. P. Chen, "Multiscale low-light image enhancement network with illumination constraint," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 11, pp. 7403–7417, Nov. 2022.

[13] R. Li, C. Duan, S. Zheng, C. Zhang, and P. M. Atkinson, "MACU-Net for semantic segmentation of fine-resolution remotely sensed images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[14] R. Zuo, G. Zhang, R. Zhang, and X. Jia, "A deformable attention network for high-resolution remote sensing images semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4406314.

[15] G. Deng, Z. Wu, C. Wang, M. Xu, and Y. Zhong, "CCANet: Class-constraint coarse-to-fine attentional deep network for subdecimeter aerial image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4401120.

[16] H.-F. Zhong, Q. Sun, H.-M. Sun, and R.-S. Jia, "NT-Net: A semantic segmentation network for extracting lake water bodies from optical remote sensing images based on transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5627513.

[17] R. Huang, C. Wang, J. Li, and Y. Sui, "DF-UHRNet: A modified CNN-based deep learning method for automatic sea ice classification from Sentinel-1A/B SAR images," *Remote Sens.*, vol. 15, no. 9, p. 2448, May 2023.

[18] S.-B. Chen, Q.-S. Wei, W.-Z. Wang, J. Tang, B. Luo, and Z.-Y. Wang, "Remote sensing scene classification via multi-branch local attention network," *IEEE Trans. Image Process.*, vol. 31, pp. 99–109, 2022.

[19] P. Lv, W. Wu, Y. Zhong, F. Du, and L. Zhang, "SCViT: A spatial-channel feature preserving vision transformer for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4409512.

[20] Z. Li, C. Yan, Y. Sun, and Q. Xin, "A densely attentive refinement network for change detection based on very-high-resolution bitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4409818.

[21] Y. Feng, H. Xu, J. Jiang, H. Liu, and J. Zheng, "ICIF-Net: Intra-scale cross-interaction and inter-scale feature fusion network for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410213.

[22] B. Yang, L. Qin, J. Liu, and X. Liu, "UTRNet: An unsupervised time-distance-guided convolutional recurrent network for change detection in irregularly collected images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410516.

[23] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[24] A. Mughees and L. Tao, "Multiple deep-belief-network-based spectral–spatial classification of hyperspectral images," *Tsinghua Sci. Technol.*, vol. 24, no. 2, pp. 183–194, Apr. 2019.

[25] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[26] Y. Ding, X. Zhao, Z. Zhang, W. Cai, N. Yang, and Y. Zhan, "Semi-supervised locality preserving dense graph neural network with ARMA filters and context-aware learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511812.

[27] Y. Ding, X. Zhao, Z. Zhang, W. Cai, and N. Yang, "Graph sample and aggregate-attention network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[28] Y. Ding, X. Zhao, Z. Zhang, W. Cai, and N. Yang, "Multiscale graph sample and aggregate network with context-aware learning for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4561–4572, 2021.

[29] X. Hu et al., "SPNet: Spectral patching end-to-end classification network for UAV-borne hyperspectral imagery with high spatial and spectral resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503417.

[30] Q. Zhu et al., "A spectral–spatial-dependent global learning framework for insufficient and imbalanced hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 11709–11723, Nov. 2022.

[31] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[32] J. Nie, Q. Xu, J. Pan, and M. Guo, "Hyperspectral image classification based on multiscale spectral–spatial deformable network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[33] W. Song, S. Li, L. Fang, and T. Lu, "Hyperspectral image classification with deep feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3173–3184, Jun. 2018.

[34] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[35] J. Yang, C. Wu, B. Du, and L. Zhang, "Enhanced multiscale feature fusion network for HSI classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10328–10347, Dec. 2021.

[36] Z. Zhang, D. Liu, D. Gao, and G. Shi, "A novel spectral–spatial multi-scale network for hyperspectral image classification with the Res2Net block," *Int. J. Remote Sens.*, vol. 43, no. 3, pp. 751–777, Feb. 2022.

[37] C. Shi, D. Liao, Y. Xiong, T. Zhang, and L. Wang, "Hyperspectral image classification based on dual-branch spectral multiscale attention network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10450–10467, 2021.

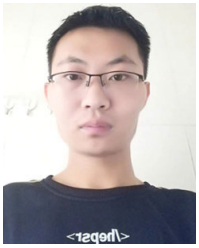
[38] S. Zhang, J. Zhang, L. Xun, J. Wang, D. Zhang, and Z. Wu, "AMFAN: Adaptive multiscale feature attention network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[39] X. Zhang, S. Shang, X. Tang, J. Feng, and L. Jiao, "Spectral partitioning residual network with spatial attention mechanism for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5507714.

[40] K. Yang, H. Sun, C. Zou, and X. Lu, "Cross-attention spectral–spatial network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518714.

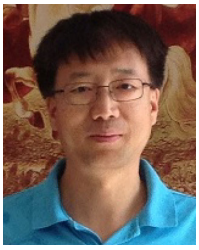
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[41] X. Zheng, H. Sun, X. Lu, and W. Xie, "Rotation-invariant attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 4251–4265, 2022.
[42] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.
[43] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[44] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," Mar. 2021, *arXiv:2103.14030*.
[45] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.
[46] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.
[47] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Aug. 2020.
[48] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, vol. 11211. Cham, Switzerland: Springer, Oct. 2018, pp. 3–19.
[49] W. Guo, H. Ye, and F. Cao, "Feature-grouped network with spectral–spatial connected attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5500413.
[50] Z. Li et al., "Hyperspectral image classification with multiattention fusion network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[51] H. Wu et al., "CvT: Introducing convolutions to vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 22–31.
[52] J. Guo et al., "CMT: Convolutional neural networks meet vision transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 12165–12175.
[53] Z. Peng et al., "Conformer: Local features coupling global representations for visual recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 357–366.
[54] Y. Chen et al., "Mobile-former: Bridging MobileNet and transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5260–5269.
[55] R. Song, Y. Feng, W. Cheng, Z. Mu, and X. Wang, "BS2T: Bottleneck spatial–spectral transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532117.
[56] J. Zou, W. He, and H. Zhang, "LESSFormer: Local-enhanced spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535416.
[57] J. Bai et al., "Hyperspectral image classification based on multibranch attention transformer networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535317.
[58] X. Yang, W. Cao, Y. Lu, and Y. Zhou, "Hyperspectral image transformer classification networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5528715.



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